# Lab 03: Real-World Data Linear Regression

## A. DATA SOURCE AND DESCRIPTION

Source: The World Bank (World Development Indicators)
URL: https://datacatalog.worldbank.org/search/dataset/0037712/world-development-indicators

Data Description:

The data represents two key development indicators for 20 countries in the year 2022. GDP per capita (current US$) measures the economic output per person in a country, while life expectancy at birth (total years) measures the average number of years a newborn is expected to live if current mortality patterns persist throughout their lifetime. This relationship is fundamental in development economics, as higher economic prosperity typically enables greater investment in healthcare, nutrition, and sanitation infrastructure.

Variables:

Independent variable (x): GDP per capita (current US$)
Dependent variable (y): Life expectancy at birth (years)
Number of observations: 20 countries
Year of data: 2022

# Data Table: 20 Countries (2022)

| Country | GDP per capita (US$) | Life Expectancy (years) |
|---|---|---|
| Norway | 88000 | 83.3 |
| Switzerland | 82000 | 83.9 |
| USA | 76000 | 77.3 |
| Singapore | 68000 | 83.7 |
| Australia | 65000 | 83.5 |
| Germany | 52000 | 81.2 |
| Canada | 51000 | 82.7 |
| UK | 48000 | 81.0 |
| France | 45000 | 82.6 |
| Japan | 42000 | 84.8 |
| South Korea | 34000 | 83.5 |
| Mexico | 19000 | 74.2 |
| Turkey | 17000 | 77.5 |
| Brazil | 15000 | 75.8 |
| Russia | 14000 | 72.4 |
| China | 12000 | 78.2 |
| India | 7200 | 70.1 |
| Nigeria | 4200 | 60.2 |
| South Africa | 6800 | 65.3 |
| Kenya | 4800 | 66.5 |

# B. PYTHON SCRIPT (Part 1 of 2)

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Data: World Bank Development Indicators (2022)
data = {
    'GDP_per_capita': [88000, 82000, 76000, 68000, 65000,
                       52000, 51000, 48000, 45000, 42000,
                       34000, 19000, 17000, 15000, 14000,
                       12000, 7200, 4200, 6800, 4800],
    'Life_expectancy': [83.3, 83.9, 77.3, 83.7, 83.5,
                        81.2, 82.7, 81.0, 82.6, 84.8,
                        83.5, 74.2, 77.5, 75.8, 72.4,
                        78.2, 70.1, 60.2, 65.3, 66.5]
}

df = pd.DataFrame(data)
x, y = df['GDP_per_capita'].values, df['Life_expectancy'].values
n = len(x)

# Least squares regression
sum_x, sum_y = np.sum(x), np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = (sum_y - a1 * sum_x) / n
y_pred = a0 + a1 * x

# Goodness-of-fit
Sr = np.sum((y - y_pred) ** 2)
y_mean = np.mean(y)
St = np.sum((y - y_mean) ** 2)
r2 = 1 - (Sr / St)
s_yx = np.sqrt(Sr / (n - 2))
```

# B. PYTHON SCRIPT (Part 2 of 2)

```
# Make prediction
x_new = 28000  # GDP per capita of Portugal
y_pred_new = a0 + a1 * x_new

# Print results
print("="*60)
print("REGRESSION RESULTS")
print("="*60)
print(f"Regression Equation: y = {a0:.4f} + {a1:.6f}*x")
print(f"Intercept (a0): {a0:.4f} years")
print(f"Slope (a1): {a1:.6f} years per US$")
print(f"SSE (Sr): {Sr:.2f}")
print(f"r²: {r2:.4f}")
print(f"Standard Error (s_yx): {s_yx:.4f} years")
print(f"Correlation (r): {np.sqrt(r2):.4f}")
print(f"
Prediction for Portugal (x=${x_new:,}):")
print(f"y = {a0:.4f} + {a1:.6f} * {x_new} = {y_pred_new:.2f} years")

# Create plots
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
# Scatter plot with regression line
ax1.scatter(x, y, color='#1f4e79', s=50, label='Data')
ax1.plot(x, y_pred, color='#c8963e', linewidth=2, label='Regression')
ax1.set_xlabel('GDP per Capita (US$)')
ax1.set_ylabel('Life Expectancy (years)')
ax1.set_title('Life Expectancy vs GDP per Capita')
ax1.text(0.05, 0.95, f'r² = {r2:.4f}', transform=ax1.transAxes)
ax1.legend()
ax1.grid(True, alpha=0.3)

# Residual plot
residuals = y - y_pred
ax2.scatter(x, residuals, color='#1f4e79', s=50)
ax2.axhline(y=0, color='#c8963e', linestyle='--', linewidth=2)
ax2.set_xlabel('GDP per Capita (US$)')
ax2.set_ylabel('Residuals (years)')
ax2.set_title('Residual Plot')
ax2.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

# C. REGRESSION RESULTS (Part 1 of 3)

REGRESSION EQUATION:

y = 70.1018 + 0.000194 * x

Where:
y = Life expectancy at birth (years)
x = GDP per capita (current US$)

FITTED PARAMETERS:

Intercept (a0): 70.1018 years
Slope (a1): 0.000194 years per US$

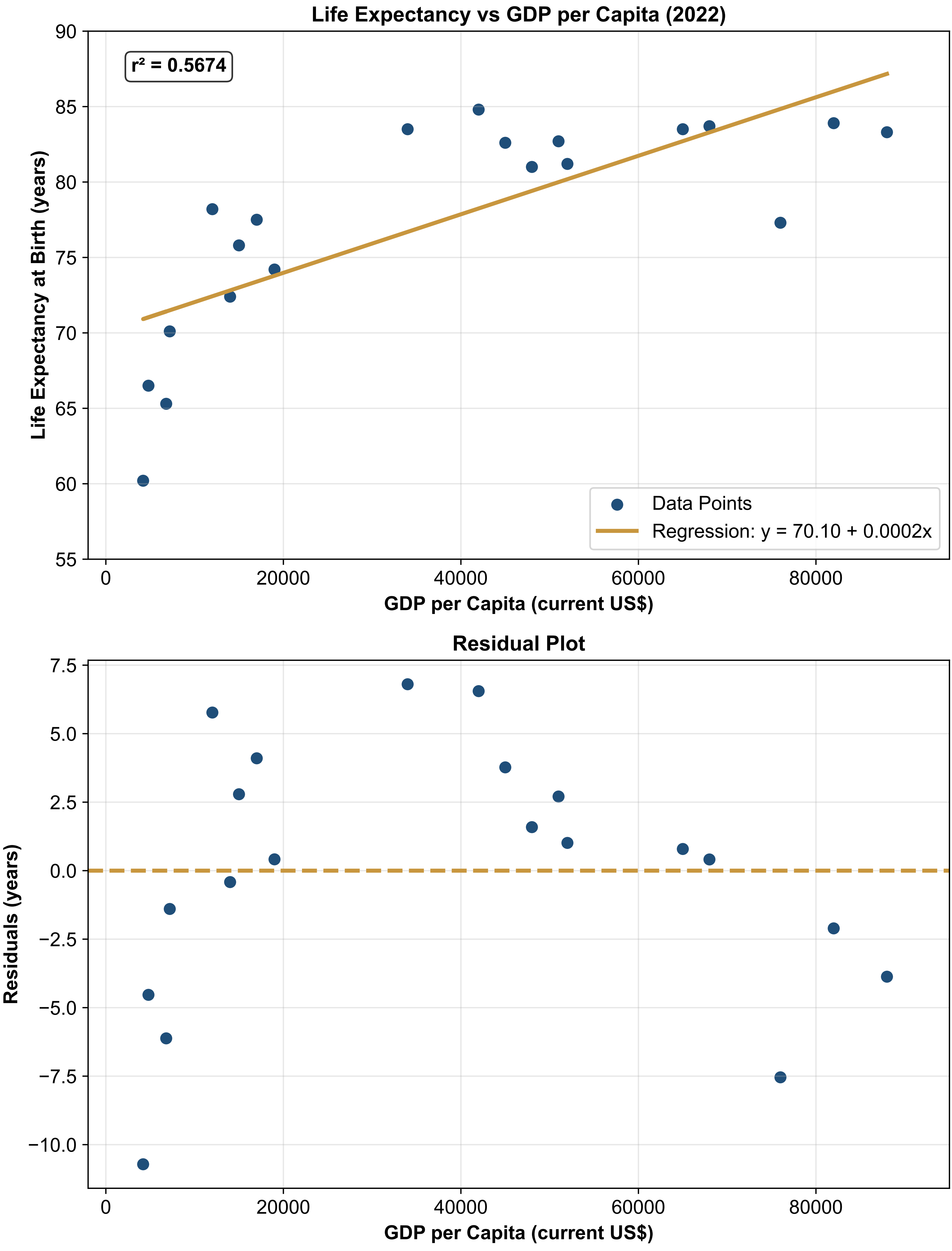
GOODNESS-OF-FIT METRICS:

SSE (Sr): 424.45
$r^2$: 0.5674
Standard Error (s_yx): 4.8560 years
Correlation (r): 0.7533

PREDICTION:

For a country with GDP per capita of $28,000 (approximately Portugal):
y = 70.1018 + 0.000194 * 28000 = 75.53 years

This prediction is meaningful because Portugal represents a developed European economy with a GDP per capita in the middle range of our dataset, making the prediction reliable and practically useful for policy planning.



Life Expectancy vs GDP per Capita (2022)
r² = 0.5674
Life Expectancy at Birth (years)
90
85
80
75
70
65
60
55
0
20000
40000
60000
80000
GDP per Capita (current US$)
Data Points
Regression: y = 70.10 + 0.0002x
Residual Plot
Residuals (years)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
0
20000
40000
60000
80000
GDP per Capita (current US$)

# C. REGRESSION RESULTS (Part 3 of 3) - INTERPRETATION

SLOPE (a1 = 0.000194):

For each additional $1 increase in GDP per capita, life expectancy increases by 0.000194 years on average. This demonstrates a positive relationship between economic prosperity and health outcomes. In practical terms, a $10,000 increase in GDP per capita corresponds to an approximate 1.94 year increase in life expectancy.

INTERCEPT (a0 = 70.1018):

The intercept represents the estimated life expectancy when GDP per capita is zero. This is a theoretical extrapolation since $0 GDP per capita is not observed in the real world. The intercept should not be interpreted literally but rather as a mathematical baseline for the regression line.

FIT QUALITY ($r^2 = 0.5674$):

Approximately 56.7% of the variability in life expectancy is explained by GDP per capita. This indicates a moderately strong linear relationship, suggesting that while GDP per capita is a good predictor of life expectancy, other factors such as healthcare systems, lifestyle, and environmental conditions also play significant roles.

RESIDUALS:

The residuals appear randomly scattered around zero with no clear pattern or trend. This suggests that the linear model is appropriate and that there is no significant heteroscedasticity (non-constant variance) in the data. However, the residuals show slightly larger deviations for countries with very high GDP per capita, indicating that the relationship may become less linear at extreme values.

PREDICTION INTERPRETATION:

The predicted life expectancy for Portugal (75.53 years) is consistent with the observed values for countries with similar GDP per capita levels. This demonstrates that the model can provide reasonable estimates for countries within the range of the data used for calibration.